

Chapter One: Cell and Its Structure

Ques. ► 1 There are two types of organic acids in organism's cell. One of them is double stranded and carries characteristics of heredity, but other one is single stranded and synthesizes proteins.
[D.B.; D.J.B.; S.B.; J.B.-18]

- a. What is centrosphere? 1
- b. What do you mean by genetic code? 2
- c. Analyze the structure of the first organic acid mentioned in the stem. 3
- d. Discuss the comparison between the two types of acid mentioned in the stem. 4

Answer to the question no. 1

a Centrosphere is the differentiated layer of cytoplasm surrounding the centriole within the centrosome.

b The genetic code is the set of rules by which information encoded in genetic material (DNA and RNA sequences) is translated into the proteins (amino acid sequences) by living cells. This code is a set of three-letter combinations of nucleotides called codons, each of which corresponds to a specific amino acid or stop signal.

c The first organic acid mentioned in the stem is DNA. The structure of DNA is described below —

The main component of chromosome is deoxyribonucleic acid or DNA. It is usually a double stranded spiral structure of polynucleotides. A strand is complementary to the other. In it there are five carbon sugars, nitrogen bases (adenine, guanine, cytosine, thymine), inorganic phosphate. These three components collectively are called a nucleotide. The Adenine of one strand bonds with the thymine of another strand by two hydrogen bonds, and guanine of a helix connects with a cytosine of another helix by three hydrogen bonds. A strand of DNA is complementary to another strand but not identical to it. A complete twist in a helix is $34\text{\AA}$ long and in a complete twist, there are ten nucleotides. So, the length between the two adjacent nucleotides is $3.4\text{\AA}$ (from top to bottom). The two strands of polynucleotides in DNA are positioned antiparallelly. It looks like rungs in a twisted ladder. The bases are connected flatly and horizontally in position from the main axis.

d The organic acids mentioned in the stem are DNA and RNA respectively. The comparison between DNA and RNA is discussed below —

DNA is a double-stranded molecule while RNA is a single stranded molecule. DNA contains the sugar deoxyribose, while RNA contains the sugar ribose. DNA has the base pairs thymine, cytosine, guanine and adenine, while RNA has the base pairs uracil, cytosine, guanine and adenine. DNA has no classification, while there are 4 types of RNA – t-RNA, r-

RNA, m-RNA, g-RNA. DNA is self-replicating, while RNA is synthesized from DNA on an as-needed basis. DNA is found in the nucleus of a cell and in mitochondria, while Depending on the type of RNA, this molecule is found in a cell's nucleus, its cytoplasm, and its ribosome. DNA is susceptible to UV damage. Compared with DNA, RNA is relatively resistant to UV damage. The number of base pair in DNA is more than RNA. DNA replicates and stores genetic information. It is a blueprint for all genetic information contained within an organism. On the other hand, RNA converts the genetic information contained within DNA to a format used to build proteins, and then moves it to ribosomal protein factories. DNA works as the main component, carrier and regulator of heredity, while RNA synthesizes proteins.

Ques. ► 2 'A' and 'B' are two organic molecule. Though 'A' can replicate itself, 'B' can not. 'B' is made with the help of 'A'.
[R.B.; C.B.; Ctg.B.; B.B.-18]

- a. What is germ cell?
- b. What do you mean by equational division?
- c. Explain the chemical structure of 'A'.
- d. 'B' is made with the help of 'A' - Explain it.

Answer to the question no. 2

a A germ cell is any biological cell that gives rise to the gametes of an organism that reproduces sexually.

b The division system, in which the identical daughter cells are produced with the same type of chromosomes in number, physical and structural features as their mother cell contains, is called equational division. Mitosis is known as equational division, because the chromosome number in each daughter cell is equal to that in the parent cell, i.e., diploid.

c "A" mentioned in the stem is known as DNA or deoxyribonucleic acid. The chemical structure of DNA is described below —

The main component of chromosome is deoxyribonucleic acid or DNA. It is usually a double stranded spiral structure of polynucleotides. A strand is complementary to the other. In it there are five carbon sugars, nitrogen bases (adenine, guanine, cytosine, thymine), inorganic phosphate. These three components collectively are called a nucleotide. The Adenine of one strand bonds with the thymine of another strand by two hydrogen bonds, and guanine of a helix connects with a cytosine of another helix by three hydrogen bonds. A strand of DNA is complementary to another strand but not identical to it. A complete twist in a helix is $34\text{\AA}$ long and in a complete twist, there are ten nucleotides. So, the length between the two adjacent nucleotides is $3.4\text{\AA}$ (from top to bottom). The two strands of polynucleotides in DNA are positioned antiparallelly. It looks like rungs in a twisted ladder. The bases are connected flatly and horizontally in position from the main axis.

d "A" and "B" mentioned in the stem is known as DNA and RNA respectively. DNA cooperates to form RNA- this is analyzed below —

The process through RNA is synthesized from DNA is known as transcription. Transcription involves three steps:

1. **Initiation:** The DNA molecule unwinds and separates to form a small open complex. RNA polymerase binds to the promoter of the template strand.
2. **Elongation:** RNA polymerase moves along the template strand, synthesising an mRNA molecule. In prokaryotes RNA polymerase is a holoenzyme consisting of a number of subunits, including a sigma factor (transcription factor) that recognises the promoter. In eukaryotes there are three RNA polymerases: I, II and III. The process includes a proofreading mechanism.
3. **Termination:** In prokaryotes there are two ways in which transcription is terminated. In Rho-dependent termination, a protein factor called "Rho" is responsible for disrupting the complex involving the template strand, RNA polymerase and RNA molecule. In Rho-independent termination, a loop forms at the end of the RNA molecule, causing it to detach itself. Termination in eukaryotes is more complicated, involving the addition of additional adenine nucleotides at the 3' of the RNA transcript.

Thus RNA is synthesized from DNA.

Ques. ▶ 3 One type of organelle of plant produce food and other type of organelle has role in the metabolism of fat and generates energy. [D.B. 17]

- a. What is fruit body? 1
- b. What do you mean by viviparous germination? 2
- c. Draw a labelled diagram of the first organ of the stem. 3
- d. Discuss the comparison between the two types of organelle mentioned in the stem. 4

Answer to the question no. 3

a The portion of *Agaricus* which is visualized on the ground is known as fruit body.

b In viviparous germination, the seeds germinate while still attached to the parent plant. The embryo grows out of the seed and then out of the fruit and projects from it in the form of a green seedling displaying the root and the hypocotyl.

c The first organ mentioned in the stem is chloroplast. The diagram of chloroplast is drawn below —

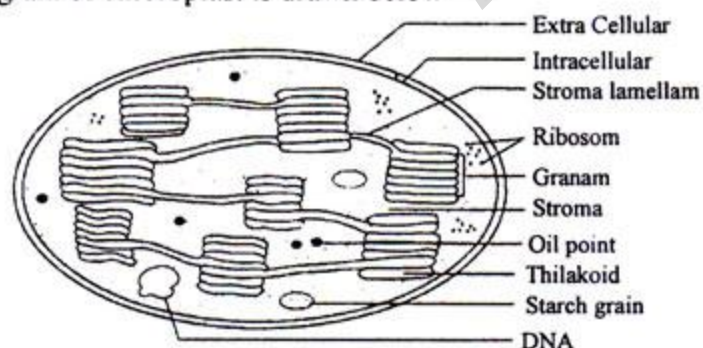


Fig- Different parts of chloroplast

d The first and the second organ mentioned in the stem are chloroplast and mitochondria respectively. The comparison between chloroplast and mitochondria is discussed below-

Chloroplasts are Usually found in plants and unicellular organisms, while mitochondria are found in almost all cells. Chloroplast has three compartments(parts): thylakoids (traps sunlight), granum (pl: grana; stacks of thylakoids), stroma (fluid inside the outer membrane, which interacts with the cytoplasm. It surrounds the granum and thylakoids. On the other hand, mitochondria has two compartments. Crista (pl:cristae) is the compartment formed by the inner and outer membrane of the mitochondria; it is the layer of folds in the

mitochondria and is studded with proteins. The other compartment is called matrix; it is the fluid inside the foldings (cristae). Chloroplasts contain pigments such as chlorophyll a, chlorophyll b and carotenoids. Mitochondria do not contain any such pigments. The ATP synthase in mitochondria and chloroplast are orientated differently. ATP synthase in mitochondria points into the matrix, with protons flowing from the intermembrane space to the matrix. In chloroplasts however, ATP synthase points towards the stroma, and protons flow from the thylakoid space into the stroma. The types of electron acceptors present in mitochondria and chloroplast vary. While mitochondria contain NAD and FAD, chloroplasts contain NADP. Chloroplasts are essential for the growth and survival of plants and photosynthetic algae. Like solar panels, chloroplasts take light energy and convert it into a usable form that powers activities. On the other hand, the most important function of the mitochondria is to produce energy. The simpler molecules of nutrition are sent to the mitochondria to be processed and to produce charged molecules. These charged molecules combine with oxygen and produce ATP molecules. This process is known as oxidative phosphorylation.

Ques. ▶ 4 A cell organelle of plant produce carbohydrate type foods. That foods are degraded by an organic system and produce energy. [R.B.-17]

- a. What is nucleic acid? 1
- b. What do you mean by genetic code? 2
- c. Draw a labelled diagram of the organ mentioned in the stem. 3
- d. Discuss the comparison between the two systems mentioned in the stem. 4

Answer to the question no. 4

a Nucleic acids are the polynucleotides having high molecular weight. The monomeric unit of which is nucleotide.

b The correspondence between specific DNA base sequence and the amino acids they specify called genetic code. This code is a set of three-letter combinations of nucleotides called codons, each of which corresponds to a specific amino acid or stop signal.

c The organ mentioned in the stem is chloroplast. The diagram of chloroplast is drawn below —

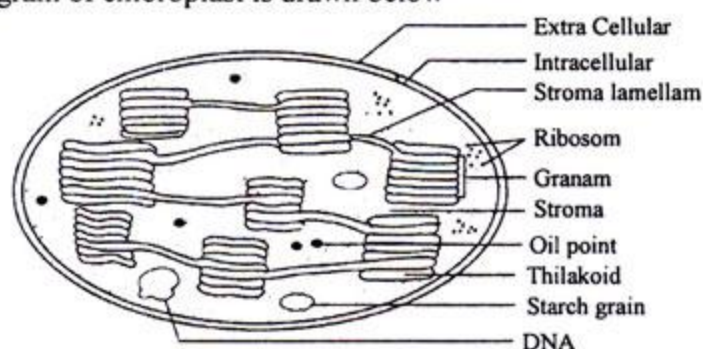


Fig- Different parts of chloroplast

d The processes mentioned in the stem are photosynthesis and respiration respectively. The comparison between photosynthesis and respiration is given below —

Photosynthesis occurs in plants whereby the plant takes sunlight and carbon monoxide from the air, and produces oxygen as a result, whilerespiration occurs in animals whereby they breathe in the air, use the oxygen to feed the body and breathe out air (containing carbon dioxide - the waste product of the body). Photosynthesis takes place in the chloroplast and is dependent on light, while respiration takes place in cytoplasm and mitochondria and is not dependent on light. In photosynthesis, light energy is fixed, whilein respiration, energy is released. Through respiration process of oxidative phosphorylation, oxidation of carbohydrates yields ATP molecules – the energy currency in living organisms,

whilethrough photosynthesis process of Photophosphorylation during photosynthesis ATP molecules are synthesized by the conversion of light energy. In respiration, glycolysis takes place in the cytoplasm and oxidation of pyruvic acid takes place within mitochondria, while in respiration, glycolysis takes place in the cytoplasm and oxidation of pyruvic acid takes place within mitochondria.

Ques. ► 5 There are two special types of organelle in plant cell. One of them takes part in respiration and the other one takes part in producing food. [S.B.-17]

- What is single membrane? 1
- What do you mean by genetic code? 2
- Describe the structure of the first organ mentioned in the stem. 3
- Analyze how the second organelle plays important role to protect the existence of the animal world. 4

Answer to the question no. 5

a Epithelial membrane of all cell organelle including plasma membrane are called single membrane.

b The correspondence between specific DNA base sequence and the amino acids they specify called genetic code. This code is a set of three-letter combinations of nucleotides called codons, each of which corresponds to a specific amino acid or stop signal.

c The first organ mentioned in the stem is mitochondria. The structure of mitochondria is described below-

Mitochondria are shaped perfectly to maximize their productivity. They are made of two membranes. The outer membranecovers the organelle and contains it like a skin. The inner membranefolds over many times and creates layered structures called cristae. ATPs are synthesized here.The fluid contained in the mitochondria is called the matrix. The mitochondrial matrix contains most the enzymes of the krebs cycle. Fatty acid oxidation also takes place here.The folding of the inner membrane increases the surface area inside the organelle. Since many of the chemical reactions happen on the inner membrane, the increased surface area creates more space for reactions to occur. Mitochondria are special because they have their own ribosomes and DNA floating in the matrix. There are also structures called granules which may control concentrations of ions.

d The second organelle mentioned in the stem is chloroplast which plays important role to protect the existence of the animal world. This is analyzed below-

Chloroplasts contain chlorophyll, which is critical for the light reactions in photosynthesis. The light reactions produce oxygen (which is expelled by the plant as a byproduct, but is critical to us). The chloroplasts also produce products that drive the Calvin cycle, which gives us carbohydrates, as a food source. Photosynthesis produces oxygen gas, which is essential for human respiration. Respiration also depends on food as a source of chemical energy. Plants use photosynthesis to make carbohydrates from water and carbon dioxide. These carbohydrates are food for many animals, which themselves become food for larger animals. All this food is then available to humans, but it would not exist without photosynthesis.If chloroplasts didn't exist, plants wouldn't have the ability to survive because photosynthesis is necessary for plant survival. And because plants provide us with oxygen, humans and animals would no longer be able to breathe. The chloroplasts also rely on humans for survival. When we breathe out carbon dioxide, the plants take it in and use it for photosynthesis. This chain of reliance among animals and plants is a major building block for sustainable life on this planet.Thus, chloroplast plays important role to protect the existence of the animal world.

Ques. ► 6

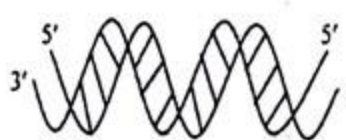


Fig: A

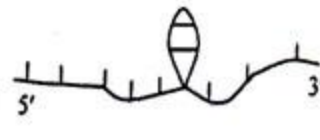


Fig: B

[D.B.-16]

- What is sorus? 1
- What is oogamous type of reproduction? 2
- Differentiate between A and B of the stem. 3
- Role of A and B are important to show the characteristics of organisms- Analyze it. 4

Answer to the question no. 6

a A sorus is a cluster of sporangia in ferns and fungi.

b Oogamous is a type of sexual reproduction in which the fusing gametes are dissimilar. One of the gametes is smaller and it behaves as the male gamete. The larger non- motile gamete acts as a female gamete. The system of reproduction between the dissimilar gametes is known as the oogamy; it is a type ofheterogamy. This type of reproduction is common in red algae and brown algae; it is also present in the green algae.

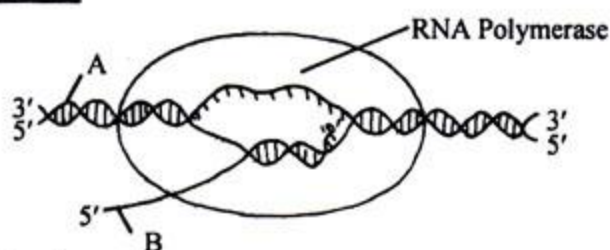
c "A" and "B" mentioned in the stem are DNA and RNA respectively. Differences between DNA and RNA are given below —

Traits	DNA	RNA
Physical structure	Double stranded	Single stranded
Chemical structure	It has two nucleotide strands which consist of its phosphate group, five-carbon sugar (the stable 2-deoxyribose), and four nitrogen-containing nucleobases: adenine, thymine, cytosine, and guanine.	Like DNA, RNA is composed of its phosphate group, five-carbon sugar (the less stable ribose), and 4 nitrogen-containing nucleobases: adenine, uracil (not thymine), guanine, and cytosine.
Types	No types	Five types- tRNA, rRNA, mRNA, gRNA and minor RNA
Replication	Replication occurs	Replication does not occur.
Location	DNA is found in the nucleus, with a small amount of DNA also present in mitochondria.	RNA forms in the nucleolus, and then moves to specialized regions of the cytoplasm depending on the type of RNA formed.
Function	DNA replicates and stores genetic information. It is a blueprint for all genetic information contained within an organism	RNA converts the genetic information contained within DNA to a format used to build proteins, and then moves it to ribosomal protein factories.

d "A" and "B" mentioned in the stem are known as DNA and RNA which play role to show the characteristics of organisms. This is analyzed below-

DNA carries all of the information for your physical characteristics, which are essentially determined by proteins. So, DNA contains the instructions for making a protein. In DNA, each protein is encoded by a gene. In DNA, each protein is encoded by a gene. RNA works along with DNA to produce the proteins needed throughout the body. During transcription, the two strands of DNA unwind. One of the strands serves as a template for make an mRNA strand. Each set of three nucleotides on an mRNA is called codon. This codons will be important in the second phase of protein synthesis called translation. After the mRNA is created during transcription, it migrates to the cytoplasm via nuclear pore. During translation, mRNA, ribosomes, rRNA, tRNA and amino acids come together to make the protein strand. These proteins show the characteristics of organisms. Thus DNA and RNA play role to show the characteristics of organisms.

Ques. ▶ 7



- What is gene?
- Which organelle is called the protein making factory of cell?
- Explain the mechanism of forming 'B' of the stem.
- Comparison between A and B of the stem.

[R.B.-16]

Answer to the question no. 7

a Gene is a specific sequence of nucleotides in DNA that is located usually on a chromosome and that is the functional unit of inheritance controlling the transmission and expression of one or more traits by specifying the structure of a particular polypeptide and especially a protein.

b Ribosome is the cell organelle which is termed as protein factory of the cell. It gathers amino acids and assembles them into protein chains to make almost anything the cell needs. A strand of ribonucleic acid, or messenger RNA, is the template. The process through which proteins are formed with the help of mRNA and tRNA is called translation.

c Process "B" mentioned in the stem is known as transcription. The mechanism of transcription is explained below —

The process through RNA is synthesized from DNA is known as transcription. Transcription involves three steps:

- Initiation:** The DNA molecule unwinds and separates to form a small open complex. RNA polymerase binds to the promoter of the template strand.
- Elongation:** RNA polymerase moves along the template strand, synthesising an mRNA molecule. In prokaryotes RNA polymerase is a holoenzyme consisting of a number of subunits, including a sigma factor (transcription factor) that recognises the promoter. In eukaryotes there are three RNA polymerases: I, II and III. The process includes a proofreading mechanism.
- Termination:** In prokaryotes there are two ways in which transcription is terminated. In Rho-dependent termination, a protein factor called "Rho" is responsible for disrupting the complex involving the template strand, RNA polymerase and RNA molecule. In Rho-independent termination, a loop forms at the end of the RNA molecule, causing it to detach itself. Termination in eukaryotes is more complicated, involving the addition of additional adenine nucleotides at the 3' of the RNA transcript.

d "A" and "B" mentioned in the stem are DNA and RNA respectively. The comparison between DNA and RNA is discussed below —

DNA is a double-stranded molecule while RNA is a single stranded molecule. DNA contains the sugar deoxyribose, while RNA contains the sugar ribose. DNA has the base pairs thymine, cytosine guanine and adenine, while RNA has the base pairs uracil, cytosine, guanine and adenine. DNA has no classification, while there are 4 types of RNA – t-RNA, r-RNA, m-RNA, g-RNA. DNA is self-replicating, while RNA is synthesized from DNA on an as-needed basis. DNA is found in the nucleus of a cell and in mitochondria, while Depending on the type of RNA, this molecule is found in a cell's nucleus, its cytoplasm, and its ribosome. DNA is susceptible to UV damage. Compared with DNA, RNA is relatively resistant to UV damage. The number of base pair in DNA is more than RNA. DNA replicates and stores genetic information. It is a blueprint for all genetic information contained within an organism. On the other hand, RNA converts the genetic information contained within DNA to a format used to build proteins, and then moves it to ribosomal protein factories. DNA works as the main component, carrier and regulator of heredity, while RNA synthesis proteins.

Ques. ▶ 8 Look at the following figures and answer the questions:

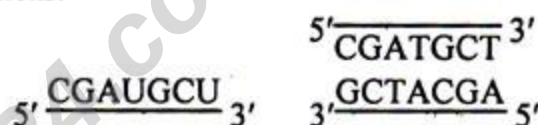


Fig: P

Fig: Q

[Dj.B.-16]

- What is mushroom?
- What do you mean by triplet codon?
- Mention the characteristics of the physical structures of fig: Q.
- Analyze the differences between P and Q.

Answer to the question no. 8

a Mushroom is an enlarged complex aboveground fleshy fruiting body of a fungus.

b The correspondence between specific DNA base sequence and the amino acids they specify called genetic code. This code is a set of three-letter combinations of nucleotides called codons, each of which corresponds to a specific amino acid or stop signal. Each of the code is called triplet codon. Each triplet codon indicates a specific amino acid.

c Fig: "Q" mentioned in the stem is DNA. The characteristics of DNA are given below —

- It contains two polynucleotide strands wound around each other. The backbone of each consists of alternating deoxyribose and phosphate groups.
- The phosphate group bonded to the 5' carbon atom of one deoxyribose is covalently bonded to the 3' carbon of the next. The two strands are "antiparallel"; that is, one strand runs 5' to 3' while the other runs 3' to 5'.
- The purine or pyrimidine attached to each deoxyribose projects in toward the axis of the helix.
- Each base forms hydrogen bonds with the one directly opposite it, forming base pairs (also called nucleotide pairs).
- 3.4 Å separate the planes in which adjacent base pairs are located.
- The double helix makes a complete turn in just over 10 nucleotide pairs. The diameter of the helix is 20 Å.

d Fig: "P" and fig: "Q" mentioned in the stem are RNA and DNA respectively. The comparison between DNA and RNA is discussed below —

DNA is a double-stranded molecule while RNA is a single stranded molecule. DNA contains the sugar deoxyribose, while RNA contains the sugar ribose. DNA has the base pairs thymine, cytosine guanine and adenine, while RNA has the base pairs uracil, cytosine, guanine and adenine. DNA has no classification, while there are 4 types of RNA – t-RNA, r-RNA, m-RNA, g-RNA. DNA is self-replicating, while RNA is synthesized from DNA on an as-needed basis. DNA is found in the nucleus of a cell and in mitochondria, while Depending on the type of RNA, this molecule is found in a cell's nucleus, its cytoplasm, and its ribosome. DNA is susceptible to UV damage. Compared with DNA, RNA is relatively resistant to UV damage. The number of base pair in DNA is more than RNA. DNA replicates and stores genetic information. It is a blueprint for all genetic information contained within an organism. On the other hand, RNA converts the genetic information contained within DNA to a format used to build proteins, and then moves it to ribosomal protein factories. DNA works as the main component, carrier and regulator of heredity, while RNA synthesis proteins.

Ques. ► 9 The teacher has drawn a labelled diagram of an ideal unit of organism's structure containing two layers. It's outer layer is dead, but inner layer is living. [Ctg. B.-16]

- What is bacteriophage? 1
- What do you mean by lichen? 2
- Draw a labelled diagram of the structure mentioned in the stem. 3
- Comparison between the two layers mentioned in the stem. 4

Answer to the question no. 9

a Bacteriophage is a group of viruses that infect specific bacteria, usually causing their disintegration or dissolution.

b The definition of a lichen is a plant found on rocks or trees made of both a specific fungus and a specific algae that help one another.

c The figure mentioned in the stem is a plant cell. A labelled diagram of a plant cell is drawn below —

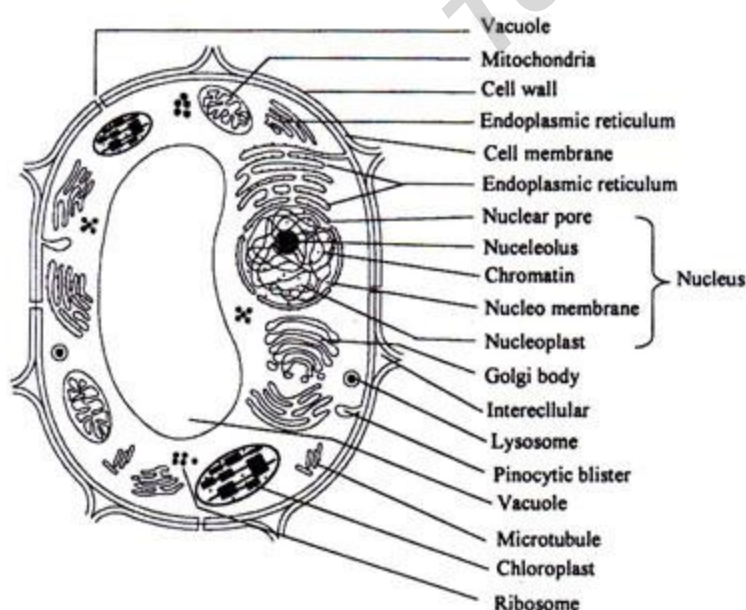


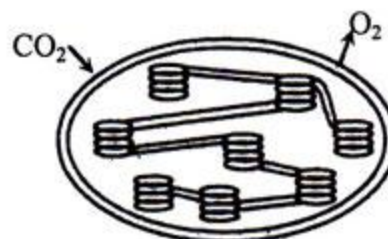
Fig: A labelled plant cell

d The outer layer mentioned in the stem are cell wall and the inner layer mentioned in the stem plasma membrane or cell membrane. The comparison between cell wall and plasma membrane is given below —

Cell wall and cell membrane both are constituents of a cell. The main difference between cell wall and cell membrane is,

Cell membrane is found in animal cells and cell wall in plant cells. Cell wall covers the cell membrane and forms the outer covering. Cell membrane is partially permeable but cell wall is completely permeable. Cell membrane do not contain cellulose, cell wall does. Cell wall is elastic, cell membrane is not. Cell wall controls turgidity of a cell, plasma membrane does not. Cell wall made up of peptidoglycan is present in a bacteria. Cell wall made up of chitin is present in fungi. Cell wall is metabolically inactive but cell membrane is metabolically active. Cell wall protects the shape of the cell and maintains it while cell membrane protect the cytoplasm and osmotic balance in a cell.

Ques. ► 10



[J.B.-16]

- What is crossbreeding? 1
- What do you mean by endosperm? 2
- Describe the structure of the stem? 3
- Analyze the importance of the system mentioned in the stem. 4

Answer to the question no. 10

a Crossbreeding is defined as the act or process of producing offspring by mating purebred individuals of different breeds or varieties.

b The endosperm is a tissue produced inside the seeds of most of the flowering plants following fertilization. It is the fusion product of secondary nucleus with one of the male gamet to form endosperm. It surrounds the embryo and provides nutrition in the form of starch, though it can also contain oils and protein. This can make endosperm a source of nutrition in animal diet.

c The organ mentioned in stem is chloroplast. The structure of chloroplast is described below-

Chloroplast is covered by 2- layered semi- permeable membrane. This is made of lipid and protein. Water absorbing matrix is covered by the membrane. This matrix is known as stroma. Numerous thylakoids are v present in the stroma. Thylakoid is sac-shaped. Some thylakoid form stack piling upon one another. This stack of thylakoid is known as granum (pl. grana). Some thylakoids of the two adjacent grana are joined together with very fine tubes. These connecting tubes are known as stroma lamelli, chlorophyll is present here too. Thylakoid membrane carries many round particles (ATP synthetase). All the enzymes for ATP production are present in it. Numerous photosynthetic units are present in these membranes. In each unit 300- 400 molecules of chlorophyll- a, chlorophyll- b, carotene, xanthophyll are found. In chloroplast there are DNA and ribosome of its own. With the help of this chloroplast cell can reproduce and synthesize some useful proteins.

d The process occurred in the stem is photosynthesis. The importance of photosynthesis is analyzed below —

The sun is the main source of all energy. Only green plants can transform solar energy into chemical energy and store as food through photosynthesis. No animal can produce its food itself. The foods we eat such as rice, bread, fruits, vegetables, fish, meat, milk, egg etc., are directly or indirectly derived from plants.

ved from green plants. So, all animals are completely dependent on green plants for their food, and the green plants produce food through the process of photosynthesis. Photosynthesis is very important for maintaining the balance of different components in the atmosphere, especially the ratio of CO_2 and O_2 . If the amount of the gases is different than normal, the atmospheric environment becomes hostile to living world. As green plants absorb CO_2 and release O_2 through the process of photosynthesis, the correct ratio of O_2 and CO_2 in the atmosphere is maintained. The progress of human civilization is dependent on photosynthesis directly and indirectly. So, if photosynthesis does not occur, human civilization would be ruined, and the living world would be destroyed. This is why photosynthesis is the most important biochemical process in the living world.

Ques. ► 11 Cell is the basic building unit of life. A typical plant cell contains one non-living outer membrane and other inner elastic, semi-permeable living membrane.

[Dhaka Residential Model College, Dhaka]

- What is plastid? 1
- Mention the functions of mitochondria. 2
- Describe the physical structure of non-living membrane. 3
- Explain the acceptable model of semi-permeable membrane. 4

Ans. to the Ques. No. 11

a Plastids are colored organelles present within plant cell which make the plant organ like leaf, flower or fruit seriously colored. They also participate in photosynthesis when green in color or store food.

b The cytoplasmic organelles covered by double layered membrane and perform special functions in respiration is known as mitochondria. The functions of Mitochondria are given below:

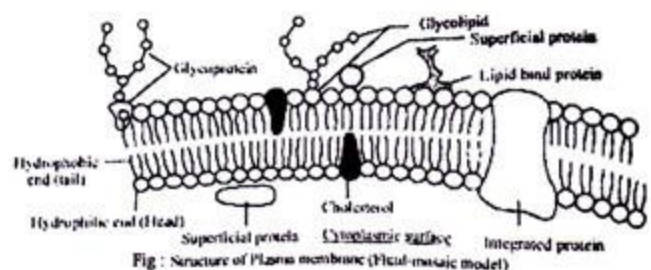
- Different stages of respiration except the glycolysis take place within mitochondria. For example: Pyruvic Acid, Krebs Cycle and Oxidative phosphorylation.
- All the respiratory enzymes remain present in the mitochondria.
- They contain all enzymes for lipid synthesis.
- Mitochondria synthesis small amount of DNA and RNA.

c The first formed cell wall remains very thin and delicate. But the matured cell wall is comparatively thick and consists of single layer. The cell walls of two literally situated cells remain attached to each other by cementing substance known as the middle lamella. The middle lamella keeps the adjoining cells attached together. The primary cell wall substances are not deposited on the middle lamella continuously but remain interrupted by some minute tube like cavities known as pits. When two such cavities remain on both sides of the cell membrane in opposite direction they are together called a Pit Pair. The middle lamella present in between the pit pair is called the pit member. The cytoplasm of the adjoining two cells remains connected by some filamentous cytoplasmic strands. These connective strands are known as plasmodesmata. The matured plant cell wall is three layered: Primary, Secondary and Tertiary. The layer of cell wall attached to the middle lamella is known as the primary wall. The layer of cell wall deposited internally on the primary wall is called secondary wall. Tertiary cell wall is formed on the inner surface of the secondary wall. The thickness of the secondary wall is not uniform all over. It becomes scattered due to thick deposition of cell wall substances following some specific patterns like annular, spiral or reticulate structures.

d Jonathan singer and Garth Nicholson in 1972 proposed the fluid mosaic model of plasma membrane. This model has gained general exceptions and reflects modern concept. According to this model, the plasma membrane is bilayered at each layer is composed of phospholipid molecules. The phospholipid molecules are bi-terminal. One end of the phospholipid molecule has hydrophilic phosphate head and the other end has a hydrophobic tail of two fatty acids. The phospholipid molecules are arranged in two opposite layers with their hydrophobic tails pointing inwards and hydrophilic phosphate heads towards the surface. Cholesterol molecules remain embedded in between phospholipid molecules. In case of animal cells, the amount of cholesterol is comparatively higher than other cells. The protein molecules remain embedded in various degrees in the two Lipid layers of plasma membrane. Carbohydrate chains may be attached to some liquid and almost all protein molecules. These are known as glyco-lipid and glyco-protein respectively. These are to combine me know as glycocalyx this works as the recognizer of cell. The carbohydrate chains always remain on external surface of the membrane, opposite to the cytoplasm.

The protein molecules remain arranged in between the phospholipid and cholesterol molecules. Depending on the arrangement in the plasma membrane protein molecules may be of three types, for example- superficial or extrinsic proteins integral protein or intrinsic proteins and lipid bind protein. The superficial proteins remain arranged on the external surface of the cell membrane. The integral proteins remain partly or fully embedded in the phospholipid layers. The lipids bind proteins remain extended through both the limitless of plasma membrane.

The membrane protein molecules are not fixed in the semi-solid phospholipid layers but move free laterally. The semi liquid nature of the plasma membrane makes the protein distributed irregularly in the phospholipid layers giving a mosaic look. For this reason this model has been named as the Fluid Mosaic Model. This model supports the presence of other Carbohydrate and protein derivatives in the plasma membrane. Carbohydrate molecules are considered to be responsible for the transition of some substances inside and Out Side the cell. Beside this, the flip flop movement of the Lipid molecules in the liquid medium is present. The result of the modern researches also support of basic model of plasma membrane.



Ques. ► 12 Observe the following flowchart & answer the questions:

DNA $\xrightarrow{A}$ mRNA $\xrightarrow{B}$ Polypeptide Chain

[Milestone College, Dhaka]

- What is virion? 1
- Why virus is called acellular particle? 2
- Write down the differences between A & B process of the stem. 3
- Analyze the importance of RNA in the process B of the stem. 4

Answer to the question no. 12

a Virion : A virion is a complete functional virus that has the capacity to infect living tissue. Viruses can be found either inside a cell or outside of a cell. If it is extracellular, the virus is called virion.

b For the following reasons virus is called a acellular Particle:

Non-living characteristics of virus: 1. Virus is submicroscopic organic particle or a cellular entity. 2. They remain dormant outside the host cell. 3. They can be crystallized by applying temperature. 4. There is no metabolic enzyme in viral body. 5. They are resistant to acid, base or salt and antibiotics have no effect on virus. 6. There is no growth of vegetative body of virus and they do not response to any stimulus. 7. Viral body consists of only a nucleic acid core and protein capsid. 8. They have no definite life period and stay in unchanged state for a very long period outside the living cell.

c The difference between 'A' transcription and B (translation) is given below:

Transcription	Translation
Transcription is the first step of gene expression process.	Translation is the second step of gene expression process.
Synthesises RNA copies of the genetic instructions written in the genome.	Synthesises proteins from RNA which are copied from genes.
Template is the genes in the genome.	Template is the mRNA.
Occurs in the nucleus.	Occurs in the cytoplasm.
RNA polymerase are the enzymes.	Ribosomes are enzymes.

d According to the stem, the process 'B' is translation process of protein synthesis. The importance of RNA in translation is given below:

Role of RNA in translation:

- (1) mRNA carries genetic information from the DNA to ribosome in cytoplasm and synthesise protein according to the encoded information with the help of tRNA and ribosome.
- (2) tRNA carries specific amino acid from the cytoplasm to the ribosome where the amino acids are used in protein synthesis.
- (3) rRNA constitute the ribosome.
- (4) gRNA: Genetic RNA function in protein synthesis and also function as the bearer and carrier of genetic characters.

Ques. ► 13 Answer the related questions from the following stem:

Singer and Nicolson introduced a model of forming plasma membrane. The model is made of two bi-layers, lipid, protein, glycocalyx and cholesterol.

[Willes Little Flower School and College, Dhaka]

- Which is called 'suicidal bag' of the cell? 1
- "Cell wall is specialised wall in plant cell"— Explain. 2
- Draw a labeled diagram of the cell membrane mention in the stem. 3
- How has the model been named? Which kind of role does it play in the organism cell? 4

Ans. to the Ques. No. 13

a Lysosome is called the suicidal bag or suicidal squared of the cell.

b "Cell wall is specialized wall in the plant cell"—

The non-living substance that remains surrounded by a limiting wall around the plant cell is known as a cell wall. No animal cell contains the cell wall.

Cell wall protects the living protoplasm from external injury, gives rigidity and mechanic strength to the cell and gives it a definite shape. As it is permeable, water and mineral salts can pass through it very easily. The cell wall maintains a connection between the living protoplasm of the adjacent cells through plasmodesmata. The cell wall plays vital role in absorption, transpiration, translocation and secretion.

Hence, cell wall is known as specialized wall in plant cell.

c

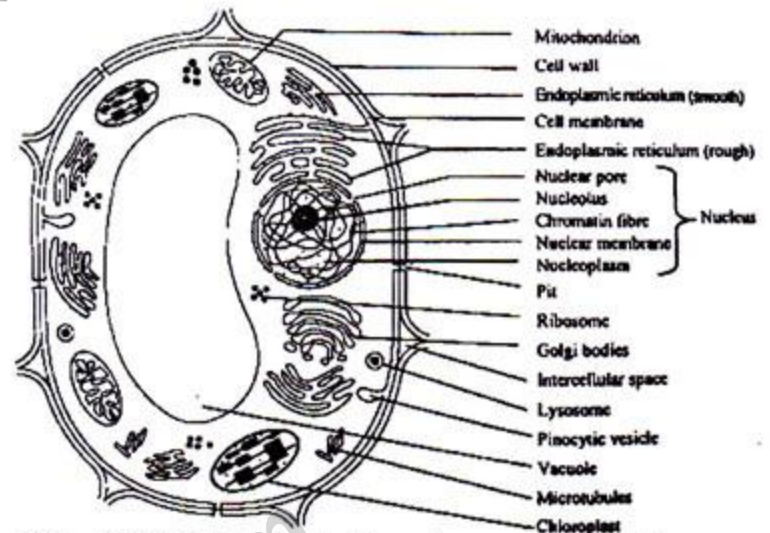


Fig-1 Labelled diagram of a typical plant cell seen under electron microscope.

d The model that has been mentioned in the passage is known as Fluid Mosaic Model and it has been introduced by Jonathan Singer and Garth Nicolson in 1972 and this model is made of two bi-layers, protein, lipid, glycocalyx and cholesterol.

The semi liquid nature of plasma membrane makes the protein inside distributed irregularly in the phospholipid layers giving a mosaic look. For this reason the model has been named as fluid mosaic model. This model supports the presence of other carbohydrate and protein derivatives in the plasma membrane. Carbohydrate molecules are considered to be responsible for transition of some substances.

The roles that the fluid mosaic model or the plasma membrane plays in organism cell are given below:

- The plasma membrane acts as a protective covering around the cell
- It is associated determination of shape of the cell
- It acts as a semipermeable membrane and controls the movement different molecules in and outside the cell
- It functions as the source of expressing different responses of cell
- It helps the cell in intaking solid materials in the process of phagocytosis and liquids in the process of pinocytosis
- It also forms some frame-works supporting specific enzymes
- It functions in forming organelles like mitochondria, golgi bodies, nuclear membrane etc.

Plasma membrane is self-sealing its wounds.

Ques. ► 14 Biology teacher told in the class that, there are two layers-A and B are present in the plant cell respectively. The A layer is dead but the B layer is live.

[Bangladesh International School and College]

- What is species? 1
- What are the importances of crossing over? 2
- Describe a well accepted model of layer B with diagram. 3
- "There are differences between the layer A and B"— Explain. 4

Ans. to the Ques. No. 14

a A biological species is a group of organisms that can reproduce with one another in nature and produce fertile

offspring. Species are characterized by the fact that they are reproductively isolated from other groups, which means that the organisms in one species are incapable of reproducing with organisms in another species.

The species is the principal natural taxonomic unit, ranking below a genus and denoted by a Latin binomial, e.g. *Homo sapiens*.

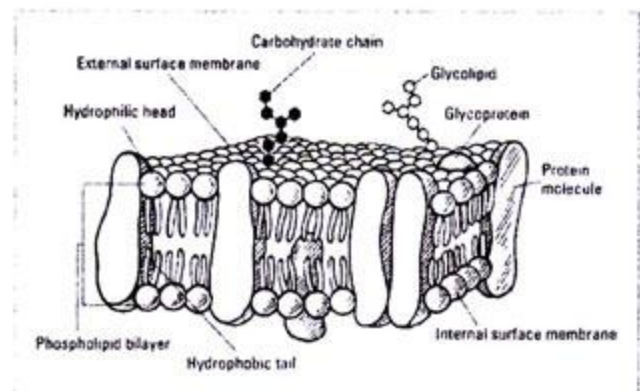
b Crossing over is the process by which homologous chromosomes exchange portions of their sequence. The importance of crossing-over-

- In the generation of genetic diversity in the resulting gametes crossing over is necessary. New genetically recombined sex cells are created. This leads to variation among offspring. Any given genetic combination can be advantageous or disadvantageous, but genetic variability of offspring increases the likelihood that at least some of the offspring will survive in an ever-changing environment.
- The frequency of crossing-over between two gene loci depends on the distance between the linked genes. Linked genes are those located on the same chromosome. Unless crossing over occurs between them, the linked genes are transferred together every time meiosis occurs. If crossing-over occurs between linked genes, alleles that are usually inherited together can be recombined and passed on separately in two different sex cells. The likelihood of crossing over between two gene loci can be used to construct gene maps.

c Layer-B in the plant cell here is the plasma membrane that is thin, elastic, living and semipermeable. The plasma membrane (also known as the cell membrane or cytoplasmic membrane) separates the interior of a cell from its outside environment. Fluid-mosaic model is the well accepted model of plasma membrane that describes the structure of the plasma membrane.

The model is described below:

- The fluid mosaic model was first proposed by S.J. Singer and Garth L. Nicolson in 1972 to explain the structure of the plasma membrane.
- The fluid mosaic model describes the structure of the plasma membrane as a mosaic of components—including phospholipids, cholesterol, proteins, and carbohydrates—that gives the membrane a fluid character.
- The plasma membrane that surrounds the cell has two layers (a bilayer) of phospholipids (fats with phosphorous attached).
- Each phospholipid molecule has a head that is hydrophilic and attracted to water and a tail that is hydrophobic and repels water. Both layers of the plasma membrane have the hydrophilic heads pointing toward the outside and the hydrophobic tails form the inside of the bilayer.
- Because cells reside in a watery solution (extracellular fluid), and they contain a watery solution inside of them (cytoplasm), the plasma membrane forms a circle around each cell so that the water-loving heads are in contact with the fluid, and the water-fearing tails are protected on the inside.
- Proteins and substances such as cholesterol become embedded in the bilayer, giving the membrane the look of a mosaic.

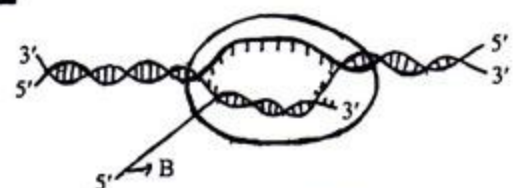


d There are differences exist between the layer A and B as we already know that layer A is dead and layer B is live. From this difference we can say that these two layers of plant cell are cell wall and cell membrane respectively. So the differences between them are:

- The cell wall is the outer most covering of the cell. The cell wall covers the cell membrane.
- The cell membrane is present in almost all types of cells. The cell wall is present in bacteria, fungi, algae and plant cell. It is absent in an animal cell and protozoa.
- The cell membrane is a biological membrane, which is semi-permeable. It allows the passage of certain substances through them and the cell wall is rigid and permeable.
- The function of the cell membrane is the same as that of the skin. It separates the components inside the cell from the outside, provides support to the cytoskeleton of the cell, gives shape to the cell, and helps in the formation of tissues by attaching the matrix found in the extra cellular. It also allows the passage of a certain substance; it maintains the potential of the cell, helps in communication with other cells, and act as molecular signals. Protein receptors are present which receives signals from the other cells and the environment. The function of the cell wall is to provide strength and rigidity to the cell. It protects the cell against mechanical forces.

The composition of the cell wall varies in prokaryotic and eukaryotic cell. In prokaryotes, the cell wall is made up of peptidoglycans in the inner layer and lipoproteins, lipopolysaccharides in the outer layer. In eukaryotes, the primary cell wall is made up of cellulose, pectin and lignin. The cell membrane is made up of proteins, carbohydrates and lipids.

Ques. 15



[Square High School & College, Pabna]

- What is gene? 1
- What do you understand by genetic code? 2
- Explain the mechanism of creation of 'B' in the above stem. 3
- Role of A and B are important in the expression of characteristics of an organism—Analyze it. 4

Answer to the question no. 15

a A unit of heredity which is transferred from a parent to offspring and is held to determine some characteristic of the offspring.

b The group of nucleotides or nitrogen bases forming the signal for an amino acid is called the genetic code. It translates information encoded within genetic material (DNA or mRNA sequences) into proteins.

c In DNA technology creation of 'B' is known as transcription. Transcription is the first step of gene expression, in which a particular segment of DNA is copied into RNA (especially mRNA) by the enzyme RNA polymerase. Both DNA and RNA are nucleic acids, which use base pairs of nucleotides as a complementary language. During transcription, a DNA sequence is read by an RNA polymerase, which produces a complementary, antiparallel RNA strand called a primary transcript.

Transcription proceeds in the following general steps:

1. RNA polymerase, together with one or more general transcription factors, binds to promoter DNA.
2. RNA polymerase creates a transcription bubble, which separates the two strands of the DNA helix. This is done by breaking the hydrogen bonds between complementary DNA nucleotides.
3. RNA polymerase adds RNA nucleotides (which are complementary to the nucleotides of one DNA strand).
4. RNA sugar-phosphate backbone forms with assistance from RNA polymerase to form an RNA strand.
5. Hydrogen bonds of the RNA-DNA helix break, freeing the newly synthesized RNA strand.
6. If the cell has a nucleus, the RNA may be further processed. This may include polyadenylation, capping, and splicing.
7. The RNA may remain in the nucleus or exit to the cytoplasm through the nuclear pore complex.

d Transcription is a vital step for expressing the characteristics of an organism. Because it is the first step in gene expression, in which information from a gene is used to construct a functional product such as a protein. The goal of transcription is to make a RNA copy of a gene's DNA sequence. For a protein-coding gene, the RNA copy, or transcript, carries the information needed to build a polypeptide (protein or protein subunit) which ultimately involves with the expression of character of an organism.

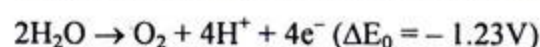
Ques. ▶ 16 On the blackboard, the teacher has drawn a labeled diagram of a unit of living organism which contains two membranes, where the outer membrane is non-living and inner membrane is living.

[The Millennium Stars School and College, Rangpur]

- a. What is photolysis? 1
- b. Why photophosphorylation is sometimes non cyclic? 2
- c. Draw the labeled diagram of the unit mentioned in the stem. 3
- d. Make a comparative discussion between the two membranes. 4

Answer to the question no. 16

a Splitting of water occurs by the action of Light and this process is called Photolysis of water or the lysis of water molecules which results in the production hydrogen and oxygen in the presence of light is called photolysis.



b The photophosphorylation process which results in the movement of the electrons in a non-cyclic manner for synthesizing ATP molecules using the energy from excited electrons provided by photosystem II is called as non-cyclic photophosphorylation.

It is thought that cyclic photo-phosphorylation is source of ATP required for chloroplast activities over and above that required in the carbon reduction cycle. Since the non cyclic photo-phosphorylation results in the production of ATP and NADPH, whereas cyclic photo-phosphorylation also represents a mechanism by which chloroplast can regulate the production of ATP/NADPH ratios, which is important in the maintenance of chloroplast metabolic activity.

c According to the stem it is a typical plant cell.

A labeled diagram of a typical plant cell is drawn below:

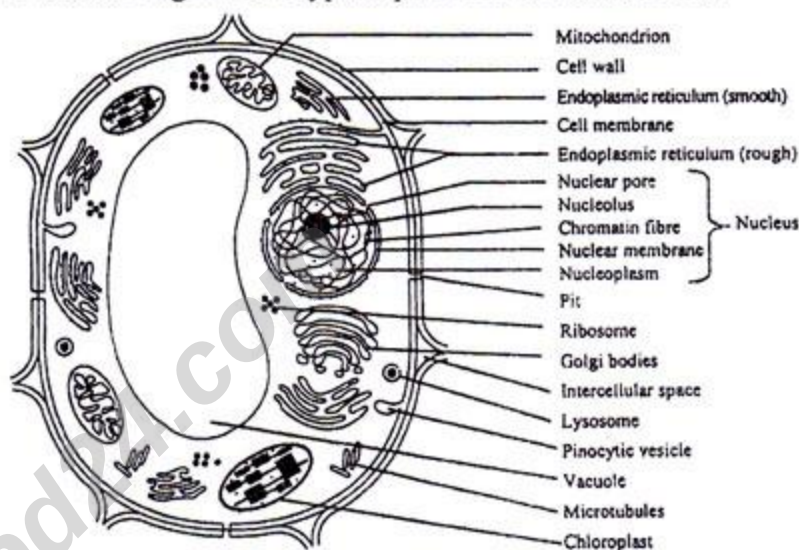


Fig: Labeled diagram of a typical plant cell seen under electron microscope.

d According to the stem it is a difference between cell wall (non-living) and cell membrane (living) :

Difference between cell wall and cell membrane.

Cell wall	Cell membrane
i. Cell wall is a non-living, rigid structure that surrounds the plant cell.	i. Cell membrane is a living, elastic, semipermeable membrane surrounding all types of protoplasts.
ii. Cell wall is mainly composed of cellulose, But, pectin, chitin etc. may also be present.	ii. Cell membrane is mainly composed of lipoprotein. But, glycolipid, phospholipid and glycoprotein may also be present.
iii. Cell wall is arranged in three layers-the middle lamella, the primary cell wall and the secondary cell wall.	iii. Cell membrane is made of two layers of protein and lipid.
iv. Pit or pore is formed here.	iv. Pit or pore is not formed here.
v. It gives a definite shape to the cell, gives rigidity and protects internal structures from external trauma.	v. Cell membrane gives shape to the cell, facilitates transport of internal organelles and increases the absorption area of the cell.

Ques. ► 17 Rafi likes to eat sweet and different oil fried tasty foods.

[Mirzapur Cadet College, Tangail]

- What is called peptide bond? 1
- What is the necessity of enzyme in human body? Explain. 2
- Explain the chemical structure of the ingredient of sweet. 3
- Analyze the harmful effects of the ingredients of oil fried tasty foods in Rafi's body. 4

Answer to the question no. 17

a A covalent chemical bond linking two consecutive amino acids is called a peptide bond.

b Enzymes play a significant role in human body. Digestion of protein, carbohydrate and fats takes place in human body. Different types of enzymes work spontaneously to do those tasks. Beside this, many physiological processes in human body are controlled by enzyme. Without enzyme those activities will be hampered.

c The ingredient of sweet is sucrose. The chemical structures of sucrose is explained below:

Sucrose (Table Sugar): The most important disaccharide in plants is sucrose. Its chemical formula is $C_{12}H_{22}O_{11}$. It is generally known as the table sugar. Sucrose is found in all green photosynthetic plants, though it is present in higher percentage in sugar cane, beet, mapple, annanas etc. The carbohydrate synthesised in leaves of green plants are translocated to different parts as sucrose. The main constituent of honey is sucrose. Sucrose is formed by the union of one molecule of glucose and one molecule of fructose.

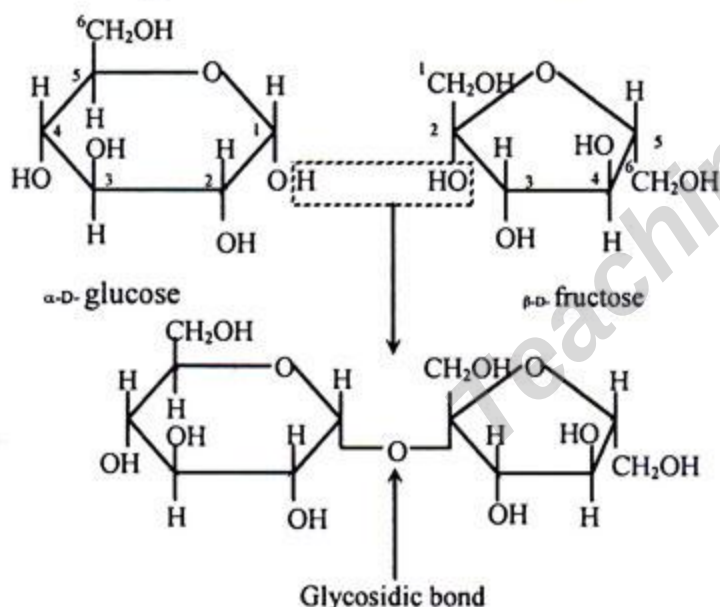


Fig: Structure of Sucrose

Sucrose is a disaccharide. Its chemical formula is $C_{12}H_{22}O_{11}$. Sucrose is formed by the union of one molecule of glucose and one molecule of fructose. OH group attached to the carbon-1 of the α -D glucose forms glycosidic bond or oxygen bridge (—O—) with the OH group attached to the carbon -2 of the β -D fructose. Sucrose is a non reducing sugar.

d Oil fried tasty foods are harmful for human body. They mainly composed of lipids or fats. These ingredients have harmful effects on health. This is analyzed below:

Oil fried foods contain fats or lipids. They are storage compounds and yield fatty acid and glycerol on hydrolysis. Consumption of excessive oil fried foods ensure intake of high amount of fatty components. This excessive fat remain deposited in human body and cause obesity. Obesity deposited

in human body and cause obesity. Obesity increases the cholesterol level. Cholesterol may be of two types. LDL and HDL. LDL is considered to be notorious cholesterol which is responsible for cardiac disorder. Increased amount of cholesterol is deposited on the wall of the blood vessels and lumen of blood vessels. It narrows down the vessel route. As a result blood pressure rises and flow of blood hampers. As a result coronary thrombosis occurs.

High consumption of oil fried tasty foods have many negative effects on human body. So Rafi should avoid these types of foods.

Ques. ► 18 DNA $\longrightarrow$ mRNA $\longrightarrow$ Protein

[Rajshahi Cadet College, Rajshahi]

- Name the pathogen of blight of Rice. 1
- Explain double fertilization. 2
- According to the given stem describe the process which occurs inside the nucleus. 3
- The process mentioned in the stem is the vital for gene expression. Explain it. 4

Answer to the question no. 18

a The pathogen of blight of Rice is *Xanthomonas oryzae*.

b During the fertilization of angiospermic plants, one sperm fuses with ovum and another fuses with secondary nucleus. This type of fertilization is called double fertilization. By double fertilization Zygote (2n) and primary endosperm nucleus (3n) is formed. Double fertilization is a unique characteristic of angiospermic plants.

c Above stem shows the transcription and translation process which produces mRNA from DNA and protein from mRNA respectively. Of these two processes, Transcription occurs inside the nucleus. The process is described below:

Transcription unit of a DNA molecule has promoted region, point of origin and termination points. At the beginning of transcription a group of proteins known as transcription factors become attached to the promoted region of the transcription unit. After this, the hydrogen bond between two complementary strands become dissolved and two strands get separated. Between these two strands one strand functions as the templet in mRNA synthesis. RNA polymerase then get attached to the promoter region. The enzyme proceeds from the point of origin in 3' $\rightarrow$ 5' direction. Newly synthesized mRNA is the pre-mRNA which contains several noncoding parts. These non-coding parts are cut out in some special mechanism and mature mRNA is formed.

The coding part of the mRNA is known as exon and noncoding part is known as introns. All the intrones are eliminated and exons are joined together by splicing. After processing is complete mature mRNA comes out through the nuclear pore and enter the Cytoplasm.

d The above mentioned processes are transcription and translation. These two processes are vital for gene expression.

Gene is the hereditary unit presents in the DNA of organism. Gene expresses a characteristic by regulating a function independently. It is a segment of DNA molecule. If a gene needs to be expressed, its information needs to be transcribed and then translated. Genes are the code sequences on a DNA

molecule. During the time of Transcription, information on DNA are transcribed to a mRNA. mRNA then become mature mRNA by splicing. Matured mRNA then translated in ribosome to produce related amino acids. Amino acids together produce protein. Protein synthesis is the ultimate aim of gene expression that means genes are expressed by synthesizing appropriate proteins. Transcription and translation play major role in protein synthesis.

The above explanation clearly states that transcription and translation processes are vital for gene expression.

Ques. ► 19

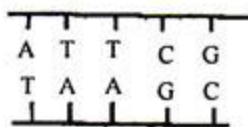


Fig-X

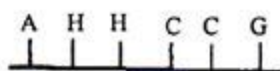


Fig-Y

[Joypurhat Girls' Cadet College, Joypurhat]

- What is codon? 1
- Which organelle is called the protein factor of cell—
Explain. 2
- Explain the difference between Fig-X and Fig-Y. 3
- Analyse the role of Fig-Y in protein synthesis. 4

Answer to the question no. 19

a Codon is a unit that consists of three adjacent bases on a DNA molecule and that determines the position of a specific amino acid in a protein molecule during protein synthesis.

b Ribosome is the cell organelle which is termed as protein factory of the cell. It gathers amino acids and assembles them into protein chains to make almost anything the cells needs. A strand of ribonucleic acid, or messenger RNA, is the template. The process through which proteins are formed with the help of mRNA and tRNA is called Translation.

c According to the stem fig : 'X' is DNA and fig 'Y' RNA. So the difference between DNA and RNA is given below :

DNA	RNA
i. DNA is a double stranded and looks like a spiral ladder.	i. RNA is single-stranded and looks like a chain.
ii. The pentose sugar is deoxyribose.	ii. The pentose sugar is ribosome.
iii. Adenine, Guanine, Cytosine and Thymine are nitrogenous bases.	iii. Adenine, Guanine, Cytosine and Uracil are nitrogenous bases.
iv. New DNA is formed by replication from DNA and by reverse transcription from RNA.	iv. New RNA is formed from DNA by transcription process.
v. DNA is the structural component of chromosome and acts as the molecular basis of hereditary.	v. The main function of RNA is to synthesis protein. But it also helps in reproduction in some viruses.

d According to the stem Fig: 'Y' is RNA. The role of RNA in protein synthesis is as follows :

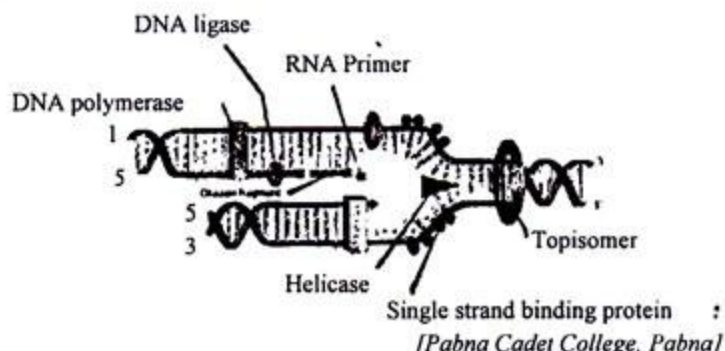
The processed RNA molecules take part in protein synthesis :

- Activation of amino acids;
- Transfer of amino acid to tRNA.
- Initiation of protein synthesis.

(iv) Elongation of the polypeptide chain.

(v) During the process of translation the genetic information's are coded in mRNA transcripts in the form of codons which in turn are specifically read by anticodon of tRNA and used to form a polypeptide molecule of defined function.

Ques. ► 20



[Pabna Cadet College, Pabna]

- What is prophage? 1
- Write down the differences between viroid and prion? 2
- Find out the role of above labeled part in mentioned process. 3
- Analyze the physical structure of an organelle where the above process is happen. 4

Answer to the question no. 20

a The attached DNA part of virus with bacterial DNA during the infection of a bacteriophage is prophage.

b The differences between viroid and prion are :

Viroid	Prion
i. RNA present in viroid : no DNA.	i. No nucleic acid present in Prion.
ii. There is no protein in viroid.	ii. Only proteins present here.
iii. Viroids are causal agents of only plant diseases.	iii. Prions are causal agents of only animal diseases.

c The above mentioned process DNA replication. Above labeled parts are DNA polymerase, DNA ligase. RNA primer, DNA helicase. Single Strand Binding Proteins and Topoisomerase. The role of these parts are as followings :

DNA Polymerase : DNA polymerase-III binds nucleotides at 5'-3' end to make complementary strand. DNA polymerase-I replaces RNA primers and fill the gaps.

DNA Ligase : Binds Okazaki segments to complementary strand.

RNA Primer : Produced by primase enzyme, be attached to the ori part.

DNA helicase : DNA helicase helps in unwinding base pairs in replication fork.

Single Strand Binding Proteins : The two strands of the DNA molecule separated by enzyme helicase are prevented from rejoining by Single Strand Binding Proteins.

Topoisomerase : Topoisomerase prevents strands from over coiling.

d The mentioned above, process happens in chromosome. The physical structure of chromosome is analyzed below :

A typical chromosome consists of different parts. Such as—

Chromatid : At the mitotic metaphase, each chromosome contain two longitudinal halves which are known as the chromatids. Each chromatid consists of a single DNA molecule.

Chromatin : Fine fibrillar bodies extended throughout the length of chromatid are known as chromatin fibre. The chromatin fibres consist of DNA and histone proteins.

Centromere : When stained, the intermediate colourless part is called centromere. Centromere region contains repetitive DNA and does not bear any gene.

Arm : The part of chromosome from the centromere to the end is called arm,

Chromomere : Beaded granular structures appear to be linearly arranged along the chromosome are known as chromomeres.

Matrix : Layer of protein and RNA below the pellicle is the matrix. It dissolves during cell division.

Pellicle : The imaginary covering of the chromosome outside the matrix is known as the pellicle.

Secondary Constriction : Some chromosome contain one or more constricted colourless parts in addition to their centromeres. These are known as secondary constrictions.

Ques. ► 21

Oreoganelle-1	Found in the nerve cell of cat and owl
Oreoganelle-2	Two types-70S and 80S

[Cumilla Cadet College, Cumilla]

- Define gene? 1
- Describe suicidal bag of a cell. 2
- How does transport occur in organelle-1? 3
- Analyze the role of organelle-2 in protein synthesis. 4

Answer to the question no. 21

a Gene is the specific part of a DNA located at a locus of chromosome which protects a specific functional gesture and comes out as a protein to grow the characteristics of organism.

b Lysosome is the suicidal bag of cell. It is like a small pouch multitudes Hydrolytic Enzyme covered with single scum. It is mostly found in animal cell and some plant cell. There are so many types of enzyme inside lysosome. Its wall bursts with severe scarcity of food and enclosed enzymes break through and destroy other microbes. This is why lysosome is called suicidal bag.

c According to the stem organ-1 is Golgi body. They are cytoplasmic organ like small capsule, blister, cistern and lamellae which are bounded by membrane near the nucleus. Golgi body contains fatty acid, vitamin, different types of enzymes. The structural variation exists in Golgi body for transport. Stacked patchwork structural shape is called cistern. In plant cell and Golgi body attaches side chain of glycoprotein and emission synthesis of complex polysaccharide. Things produced in endoplasmic reticulum accept the part of a membrane vesicle through medial cisterna ultimately transisternae can be seen in its where in the cell or plasma membrane. Besides, synthesis of non-protein, emission of enzyme, removal of cell water, is also carried out by the Golgi body.

d The organ mentioned in the stem is Ribosome. Ribosome plays an important role in synthesis of protein. Importance of Ribosome in protein synthesis is analyzed below. Protein synthesis starts from DNA through the creation of mRNA. At the beginning of protein synthesis mRNA original cell 30s and actual cell 40s connects with sub-unit. Then 30s with 50s and 40s with 60s sub unit come together to form the 70s (original cell) and 80s (actual cell) units. After that Amino Acid chain

produced in the code named mRNA. Ribosome is used here as the place of work. The ribosome contains two sites of the big one. A site and P site. These Ribosome sites create polypeptide chain. The important step of protein synthesis occurs at ribosome that is why ribosome is called protein factory of cell.

Ques. ► 22 Teacher was discussing about protein synthesis. For protein synthesis the first step is the production of mRNA. The next step is occur in cytoplasm.

[Cumilla Cadet College, Cumilla]

- What is amino acid? 1
- Describe about replication fork. 2
- How does the product from first step finalized from preliminary stage. 3
- Analyze the role of RNA for the second step. 4

Answer to the question no. 22

a One or more hydrogen atom of an organic acid which is replaced by amino group ($-NH_2$) called amino acid.

b To begin the replication process, DNA releases the alkaline joint from one or more place and creates the origin. The atom of initial protein connects with the original part to create a compound. It opens a DNA ring.

For that the alkaline joint opens up from the both side of original area and starts moving forward which produce Y shaped replication fork. If the transcription of DNA atom starts on one edge, replication fork would be one faced and if it occurs on the middle edge, the fork would be two faced.

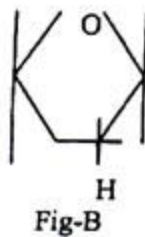
c The first step of protein synthesis mentioned in the stem is making mRNA.

It is produced in the initial stage and becomes ready to protein synthesis at the last stage. The process of its initial stage to reach the last stage is described here-

RNA polymerase enzyme connects with the promoter part of transcribing formula. In the new mRNA, transcribing sequence is being made against of DNA alkaline. Just Uracil happens instead of DNA thymine. When RNA polymerase reaches the end point, mRNA gets free. Newborn mRNA is called pre-mRNA. The signaling part of pre-mRNA is called axon and the hintless part is intron. Introns are being cut off and axons are being joined up with the help of enzyme. It is called gene splicing. In this way pre-mRNA turns up into matured mRNA and comes up to cytoplasm from nucleus.

d The second stage mentioned in the stem is translation process. Protein synthesis occurs from mRNA in this process. In the mean time, different RNA plays an important role. The role of RNA in translation is described here-

The first condition in translation is to transfer the message from DNA to RNA. It is mainly done by mRNA. Later mRNA is translated to ribosome. The whole process of protein synthesis is mainly one by the combined work of different RNA. The producing process of protein wouldn't be possible if the RNA were non-working. If RNA does not connect to mRNA in the translation, then it is not possible to produce protein. Again, if mRNA and RNA formulas don't make complementary base pairing in the opposite faced place, the polypeptide won't be made as well. Now it is clear from the discussion that different RNA plays a vital role in protein synthesis.

Ques. ► 23

[Feni Girls' Cadet College, Feni]

- What is capsid? 1
- How pollen is produced? 2
- Describe the production of molecule with A from DNA for protein synthesis. 3
- Explain the significance of nucleic acid molecule with B in creating variety. 4

Answer to the question no. 23

a Capsid: The protein coat of virus, which is made up of units called capsomeres. The chemical nature of the capsid is important in stimulating the body's immune response against the invading virus.

b Pollen Formation:

In coniferous plants, pollen is formed in the microsporangia of the male cone, whereas it is produced in the anthers of flowering plants (also termed angiosperms). Each microsporocyte is diploid and forms four haploid cells, termed microspores, via meiosis. This process is termed microspogenesis. The four microspores then form the double wall of the pollen grain within a structure made of callose. During this process, the callose is digested by callase and the pollen grains are able to grow and complete the formation of the endosperm and exosperm.

c According to the stem fig: A represents the pentose sugar which is called ribose sugar. The nucleic acid containing ribose sugar is called ribonucleic acid or RNA. The functions of RNA for protein synthesis are as follows:

Function of RNA in protein synthesis:

- mRNA carries genetic information from the DNA to ribosome in cytoplasm and synthesizes protein according to the encoded information with the help of tRNA and ribosome.
- tRNA carries specific amino acid from the cytoplasm to the ribosome according to the encoded information of the mRNA.
- They function in protein synthesis and also function as the bearer and carrier of genetic characters.

d According to the stem fig: B is β -D Deoxyribose. Significance of β -D deoxyribose is given below:

Function of DNA for creating variety:

Functions of DNA:

- It functions as the structural component of the chromosome.
- It functions as the molecular basis of heredity.
- It bears the heredity characters of organisms and transmits the characters to the next generation.
- It regulates all the activities of the organism through protein synthesis.
- It creates variation of character by undergoing mutation which is the raw material for evolution.

- It is responsible for the expression of all characteristics of the organism.
- It repairs any fault created in the structural frame of DNA molecule.

Ques. ► 24 Observe the stem and answer the question below :

[Faujdarhat Cadet College, Chattogram]

- What is hypha? 1
- What do you mean by triple fusion? 2
- Describe the process A mentioned in the stem. 3
- DNA does not express characters without process B—explain. 4

Answer to the question no. 24

a The body of multicellular fungi consists of narrow cylindrical filaments which may be repeatedly branched. Such a fungal body consisting of aggregated mass of filaments is called mycelium and each filament of the mycelium is called hypha.

b The second spore reaches the secondary nucleus and they fuse to form the primary endosperm nucleus. Triploid endosperm is produced from it. This fusion of the spore with the secondary nucleus is called triple fusion.

c According to the aforementioned paragraph, A is identified by the process of DNA to RNA. The process of transferring the encoded information from the DNA to mRNA is called transcription.

The process of this stage is given below :

- In the action of RNA polymerase enzymes, the fragments of double helix are broken by breaking down the hydrogen bonds of a specific part of the DNA molecule. As a result, the nitrogenous bases of each part of this part of the DNA molecule are exposed.
- Each nitrogenous base exposed to the 3'-5' of the DNA molecule of the DNA molecules attracts its complementary RNA nucleotide, which means that the DNA is exposed to the open nitrogenous cytosine, which is associated with RNA nucleotide and adenine-containing uracilic RNA nucleotide.
- The RNA polymerase enzymes form a single chain by DNA molecules moving along a specific direction to break the double helix and with the help of Mg^{2+} or Mn^{2+} , the continuous active complementary RNA nucleotide with each nitrogenous base exposed to the DNA molecule 3'-5' of the DNA molecules.
- Due to the addition of complementary RNA nucleotide by separating double helix at specific points, the same enzyme helps in the reconstruction of the DNA molecule on its opposite side. The density of molecule DNA molecule 3'-5' is used in the form of mRNA.
- At the same time 12 RNA nucleotide is formed with mRNA chains, and RNA polymerase enzyme forms very numerous mRNA chains from different regions of the DNA molecule.

6. The synthesized mRNA molecule goes out of the nucleus to the cytoplasm through a nucleus hole in the DNA molecule and goes into the protein synthesis site ribosome.

The synthesized mRNA molecule is composed of supplemental alkaline made from the area of the DNA molecule. In the case of mRNA uracil is used instead of thymine as supplementary of adenine. Besides, ribose carbs are used instead at deoxyribose.

d DNA is the holder and carrier of the family's hereditary trait. The main function of DNA is to express the characteristics of the organisms. The genes are located in DNA. through the jinn, the characteristics of the organism are manifested and they are transmitted in progeny. Gene is a documented piece of DNA that normally contains the formation of polypeptide chain and all the information about the control is preserved. Various features of jib such as the color of the shin, the hair properties, etc are visible in those special polypeptides, which results in protein synthesis.

In the aforementioned paragraph the process B is translation. The need for gene expression in organisms is to create RNA through transcription from the genes portion of DNA and to produce polypeptide chain (protein) from mRNA, which is called a translation, which is identified by B. Translation is the translation of DNA into protein languages through mRNA. According to the properties of DNA are expressed in the form of proteins produced by the translation. Various essential and essential enzymes are formed in combination with this protein. This enzymes release all the vital chemical properties of the organism as well as external properties of the organism. If the translation process is not done then the properties stored in the DNA will not be converted into proteins, hence the corresponding properties of the organism will not be released. So, it can be said that the properties of DNA will not be published without process B mean translation.

Ques.► 25 A- A process through which RNA is produced from DNA

B- A process through which protein is produced

[Sylhet Cadet College, Sylhet]

- What is Lysosome? 1
- What do you mean by inflorescence? 2
- Analyze the structure of an organelle which is related with the mentioned process in "B". 3
- Explain the initiation step of above process "B". 4

Answer to the question no. 25

a Lysosome is a small sac like organelle surrounded by a unit membrane which consists of several hydrolytic enzymes.

b Inflorescence is the mode of arrangement of flowers on the floral axis. The arrangement of old and younger flowers is shown by the inflorescence. Inflorescence is an unique character to identify plants, Racemose, Cymose, Spikeletes are some types of inflorescences.

c The mentioned process in 'B' is the Translation process by which protein is produced. Ribosome is the main organelle related with the Translation process. The structure of ribosome is analyzed below:

Ribosomes are present in all types of cells including Prokaryotic and Eukaryotic cells. Ribosome is membraneless organelle. Its diameter is about 22nm and height is about 20

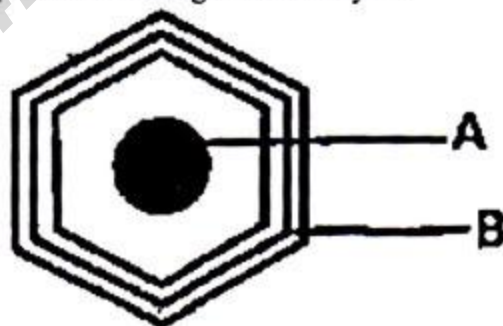
nm. All types of ribosomes consist of two unequal sub-units. Mainly two types of ribosomes such as 80s and 70s are present. 80s ribosome consists of two subunits of 60s and 40s. 70s ribosome consists of two sub units of 50s and 30s. Ribosome contains mainly two chemical components, eg protein and rRNA. A complete 70s ribose contains about 50 types of protein and 3 types of rRNA. It also contain RNAase enzymes and some metallic ions, eg. Mg^{2+} , Ca^{2+} and Mr^{2+} .

d The above mentioned process in-B through which protein is produced is Translation. The initiation step of the Translation process is explained below:

The different amino acids participating in protein synthesis remain present in the cytoplasm and become activated by the addition of ATP with the help of enzyme amino-acyl synthetase and then attached to the free 3' OH end of the corresponding tRNA molecule. mRNA coming from the nucleus become attached to small sub-unit of the ribosome. There are two sites of in the ribosome for tRNA. Initiation code of mRNA when comes in 'A' site, the tRNA bearing methionine becomes attached to the mRNA by its anticodon part. At this time the larger sub-unit of the ribosome becomes attached to the small subunit and forms a initiation complex. mRNA gradually moves from 'A' site to 'P' site in 5'→3' direction.

Thats how the initiation step of the translation process starts.

Ques.► 26 Observe the figure carefully—



[Jhenaidah Cadet College, Jhenaidah]

- What is peptide bond? 1
- Why glucose called aldohexose? 2
- Describe the features of polysaccharide found in B. 3
- If A goes to reductional division each daughter cell will be different— Explain with diagram. 4

Answer to the question no. 26

a A peptide bond is one type of covalent bond linking two consecutive amino acids.

b Glucose is an important monossaccharide.

Monossaccharide containing 6-carbon atoms are called hexosle. Glucose contains 6-carbon atoms and its chemical formula is $C_6H_{12}O_6$. It also contains aldehyde group. Thats why glucose is called an aldohexose.

c The polysaccharide found in 'B' is cellulose. The features of cellulose are described below:

Cellulose is the most important and abundant polysaccharide in the plant kingdom. Approximately 1250 to 12,500 α and β glucose molecules linked in 1-4 glycoside bond form the linear cellulose molecule which are hydrogen bonded along their lengths with other similar chains to form the cellulose fibre. It is a tasteless, odourless tough polysaccharides. It is insoluble in water and organic solvents. It is generally chemically inert

substance without reducing capacity. It produces glucose molecules when hydrolysed with concentrated acid solutions. Cellulose is the main constituent of textile fibres, papers, photo films, packaging materials etc. It is also used as the main component of cattle fields.

d 'A' mentioned in the above stem is nucleus. During cell division reductional division of nucleus gives rise to daughter cells which are different than the mother cell. The total process is explained with diagram below:

Reductional division of cell is the meiosis cell division. In meiosis, a diploid ($2n$) cell divides in two steps through some intermediate phases and produce four haploid cells. As a result chromosome number becomes half in the daughter cells.

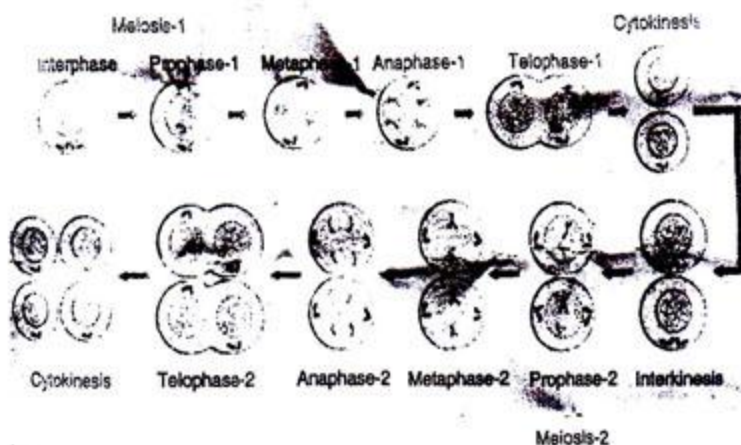


Fig : Meiosis cell division

In meiosis cell division, crossing over between non-sister chromosomes occur. As a result characteristics rearrangement occurs and daughter cells become different.

Above discussion clarifies that if reductional division occurs in nucleus, each daughter cell will be different.

Ques. ▶ 27 Rofiq said to his younger brother Sofiq who is a student of biology that there are two layer 'A' and 'B' in the plant cells. The outer layer 'A' is non living and the inner layer 'B' is living.

[Barishal Cadet College, Barishal]

- What is plasmogamey? 1
- Why lysosome is called suicidal bag? 2
- Draw labeled diagram of universally accepted model of the layer 'B' mentioned in the above stem. 3

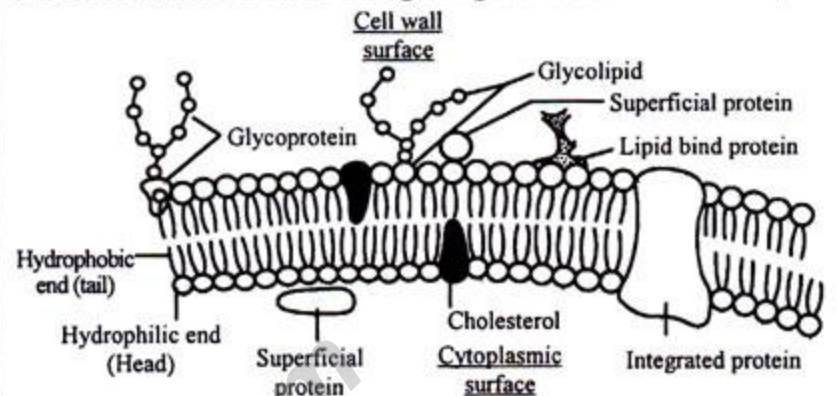
d. Analyse the difference between the two layer A and B. 4

Answer to the question no. 27

a Two genesis cells of fungus of protoplasm union is known as plasmogamy.

b In the lysozyme, there are varieties of enzymes. Sometimes, the wall of the intense food bursts and covered enzyme get out of the house and others small of cell were destroyed. That's why lysosome is known as suicidal squad.

c In the stem B is the layer of cell membrane. The universal model of cell membrane's diagram given below —



d In the stem the layer A is cell wall and layer B is cell membrane. Between A and B there are so many differences. The cell wall of plant cells has many properties, but there are no properties in the animal cell. On the other hand, both of them have plasma membrane. The cell wall is mainly made of hard, inert wall. Again the cell membrane is made of protein and lipid which have life, elastic and layer. The cell wall is arranged by three layers. They are middle layer, primary and secondary layer. But the middle stage of cell membrane has two layers. They are arranged by lipid on the outside and inner side has protein. The cell wall is lifeless but cell membrane has life. The main work of wall is shape-controlling and firmness of cell given. The main work of cell membrane is inner and outer side of cell controlled and collected protoplasm of cell. The cell wall has "olomkoron" but there is not "olomkoron" in the cell membrane.

The cell wall has three layers and the cell membrane has two layers.